

Athish Perumal

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Full-stack Developer with **2+ years of experience** building applications using **Node.js, Next.js, AWS and PostgreSQL**. Experienced in **REST APIs, GraphQL, database design, cloud deployment, and authentication**.

EXPERIENCE

Software Engineer Intern, Holiday Channel

Jul 2026 – Present

- Revamped the Budget Tracker using **React/TypeScript, RTK Query, Node/Express, and Supabase/PostgreSQL**, consolidating competing backends, removing **3000 lines of legacy code**, and implementing live CRUD for budgets and expenses.
- Resolved **5+ database and authentication issues** including schema permissions, UUID/FK mismatches, cookies, and session handling. Rebuilt auth with self-healing Auth0 user provisioning.
- Reduced API response times by **95% (10.4s → 0.25s)** by fixing a global database retry bottleneck. Integrated the **Stripe API** to support secure payment processing, handling checkout flows, payment status, and transaction data within the application.

Software Engineer, AAtek

Jul 2022 – Jun 2024

- Built an admin console and robot monitoring platform that provides real-time visualization of telemetry including status, temperature, torque, cycle time, and performance across **3+ client environments** using **Next.js**.
- Designed the backend for telemetry ingestion, robot health monitoring, alerting, and maintenance workflows, exposing a **GraphQL API** for querying device status, telemetry history, and alerts. Optimized **PostgreSQL queries** and implemented **Redis caching**, reducing API response latency for frequently accessed robot status endpoints by **70%** and improving dashboard load time by **50%**.
- Integrated an **event-driven backend using SQS** for fault-tolerant telemetry processing. Containerized and deployed services with **ECR and ECS on AWS**.

Software Engineer, Velocita Racing

Jul 2021 – Jun 2022

- Maintained a **React.js + Tailwind CSS** team website and developed a **Node.js inventory management system** for team operations and asset tracking.
- Migrated website hosting to **AWS CloudFront and RDS**, reducing page load time by **40% (2.5s → 1.5s)** and improving website delivery performance and reliability.

PROJECTS

Slotora – Multi-Tenant Appointment Booking Platform

- Engineered a **multitenant appointment booking SaaS** using **Java 21, Spring Boot, React, PostgreSQL, and JPA/Hibernate**; designed a **5-entity schema**, JWT authentication, and a **3-step business owner onboarding flow**.
- Built **19 automated tests with 0 failures** using **JUnit 5, Mockito and Testcontainers**; containerized with **Docker Compose** and deployed the full stack using **AWS Elastic Beanstalk, RDS, S3, and CloudFront**, with environment-based credential management.

Roommate Harmony – Full-Stack Household Coordination Platform

- Built a group-scoped household coordination platform for **expense splitting, chore scheduling, hangout planning, and issue tracking** using **React 18, Vite, Tailwind CSS, Shadcn UI, Node.js, Express.js, Prisma ORM, and PostgreSQL/Supabase**, with a strict **Controller → Service** backend architecture.
- Implemented **JWT authentication with HTTP-only cookies, bcrypt, CORS, and React Router v6 route guards**; added **Supertest API integration and Playwright E2E testing**, and deployed the full stack using **Vercel, Render, and Supabase**.

Cortex – Document Intelligence & RAG Platform

- Engineered a **RAG service** using **FastAPI, PostgreSQL/pgvector, SQLAlchemy, Voyage AI, and Gemini**, implementing document ingestion, recursive chunking, batched embeddings, cosine-similarity retrieval, and source-based responses with **1024-dimensional vector embeddings**.
- Designed an asynchronous **routes → services → database** architecture; containerized with **Docker**, deployed using **Kubernetes**, and implemented **GitHub Actions CI/CD** for automated testing, image builds, and deployment workflows.

EDUCATION

University of North Carolina at Charlotte

Aug 2024 – May 2026

Master of Science in Information Technology

GPA: **4.0/4.0**

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL, HTML/CSS **Frontend:** React.js, Next.js, React Router, Vite, Tailwind CSS, Shadcn UI

Backend & APIs: Spring Boot, FastAPI, Node.js, Express.js, REST, GraphQL, JWT, JPA/Hibernate, Prisma, SQS

Infrastructure: PostgreSQL, pgvector, MongoDB, Redis, AWS (ECS, ECR, RDS, S3, CloudFront), Docker, Kubernetes

Testing CI/CD: JUnit 5, Mockito, Supertest, Playwright, GitHub Actions.